National University of Computer & Emerging Sciences - NUCES - Karachi
FAST School of Computing

Artificial Intelligence AI-2002

AI Lab 4 Tasks

Task 01

Implement a binary tree of alphabets (26 nodes) where each node is initialized randomly with an alphabet. Implement the following uninformed search algorithms to find the goal(G) node.

a) Breadth First Search

b) Depth First Search

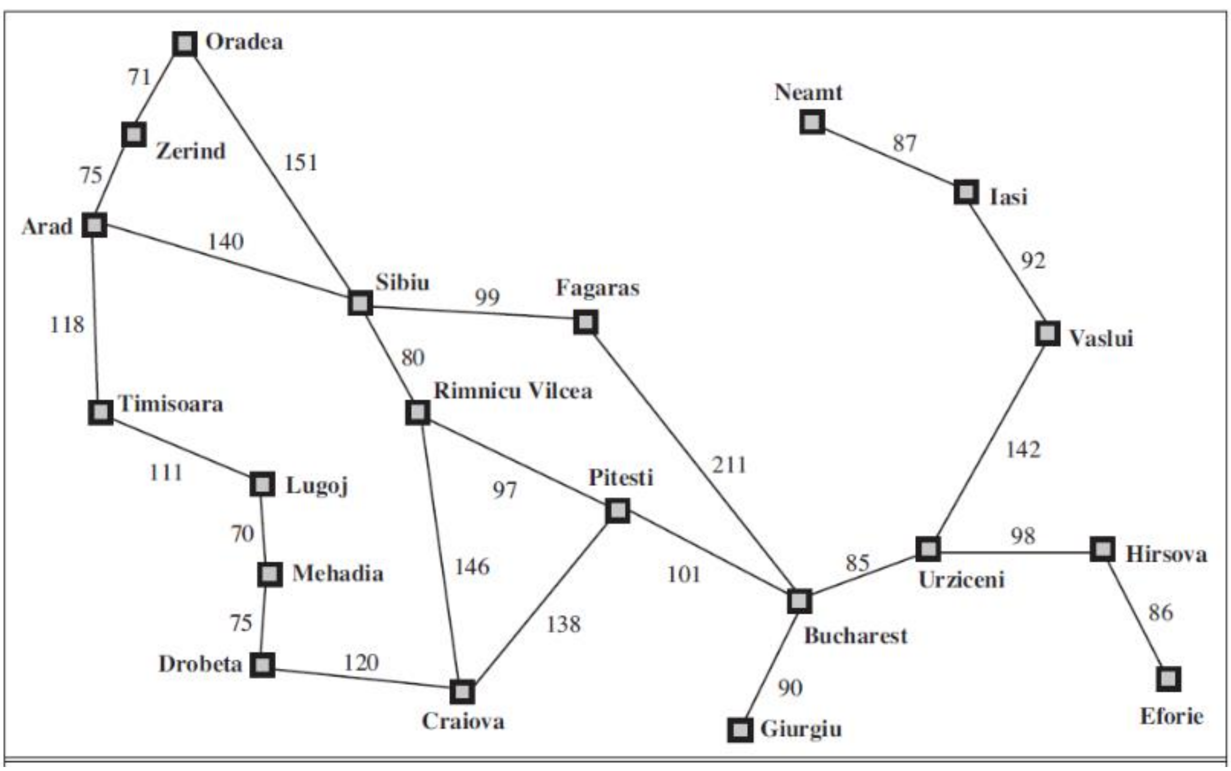
c) Depth Limited Search

d) Iterative Deepening Search

e) Uniform Cost Search

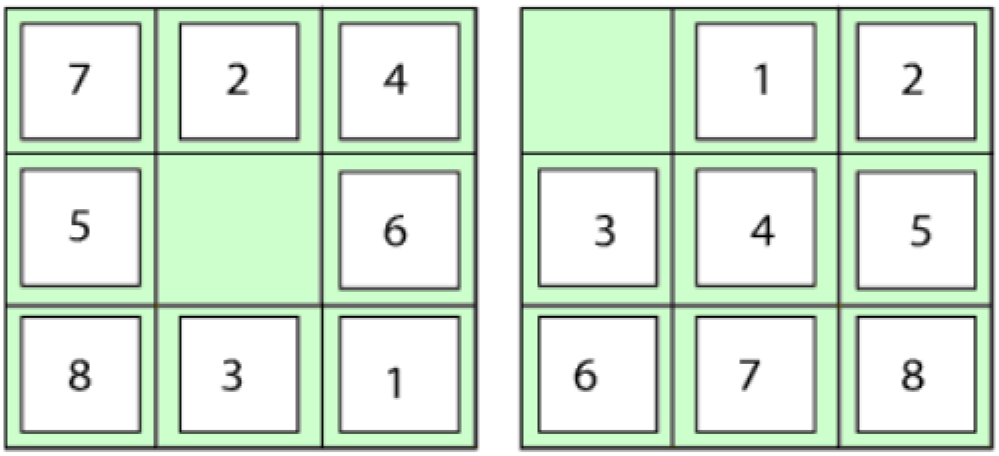
Task 02

Implement any of uninformed search algorithm to find the path from initial state to the goal state. Show the results by considering initial(Arad) and goal(Bucharest) states. Provide the reason for the choice of algorithm.



<u>Task 03:</u> 8 Puzzle Problem

Here, we have a **3x3 matrix** with movable tiles numbered from **1 to 8 with a blank space.** The tiles adjacent to the bland space can slide into that space. The objective is to reach specified goal state.



Start State　　　　Goal State

Implement an agent and show the final state.